

MARNITZ, Germany

WMO: 102640

Lat: 53.3222N

Lon: 11.9319E

Elev: 286

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.7	13.1	2.2	6.2	10.3	6.8	7.9	13.8	24.5	42.7	21.7	40.8	3.5	70	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	83.2	66.0	79.9	64.3	76.5	63.1	67.8	79.0	66.0	76.8	64.2	73.9	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	89.5	71.8	61.8	83.4	70.2	60.1	78.5	68.7	32.3	79.1	30.8	76.4	29.5	73.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.2	16.2	DB	6.9	88.7	11.4	2.4	-1.3	90.5	-8.0	91.9	-14.4	93.2	-22.7	95.0	(4)
(5)			WB	6.5	71.6	10.8	1.6	-1.2	72.8	-7.5	73.7	-13.6	74.6	-21.4	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	32.5	32.0	39.2	45.6	54.2	58.9	63.1	63.2	56.3	48.8	39.8	35.5		
	DBStd	13.07	10.53	9.02	7.02	6.48	6.49	5.95	5.30	5.20	4.89	6.46	6.20	7.02		
	HDD50	2444	543	504	339	162	32	1	0	0	5	99	308	451		
	HDD65	6491	1008	923	800	581	338	201	102	99	263	503	757	916		
	CDD50	1542	0	1	4	32	163	270	405	409	195	61	2	0		
	CDD65	113	0	0	0	0	4	19	42	44	3	1	0	0		
	CDH74	1244	0	0	0	3	90	245	486	380	39	1	0	0		
	CDH80	255	0	0	0	0	6	59	107	80	3	0	0	0		
	WSAvg	7.5	9.2	8.5	8.4	7.4	6.4	6.2	6.3	5.9	6.8	7.2	8.2	9.0		
	PrecAvg	24.4	2.0	1.7	1.6	1.3	2.0	2.5	2.8	2.6	2.0	1.9	1.9	2.1		
(15) Precipitation	PrecMax	33.4	4.1	4.1	3.8	2.9	4.6	4.7	6.3	5.3	5.8	4.4	4.1	4.4		
	PrecMin	17.6	0.0	0.3	0.3	0.1	0.4	0.6	0.7	1.0	0.5	0.8	0.2	0.5		
	PrecStd	4.0	1.1	0.8	0.9	0.7	1.0	1.1	1.4	1.1	1.2	0.9	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	57.2	62.1	71.7	80.5	86.2	86.5	86.4	78.5	70.6	55.4	52.9		
		MCWB	48.4	51.3	51.4	56.6	61.3	66.4	67.3	68.3	63.7	61.3	51.5	49.5		
	2%	DB	48.6	51.2	56.1	65.6	76.6	80.4	83.2	82.1	73.7	64.8	52.8	50.7		
		MCWB	45.9	47.0	48.7	53.1	60.4	63.6	66.2	65.7	62.9	57.7	50.2	48.2		
	5%	DB	46.5	46.6	51.7	61.2	72.2	75.9	80.0	78.5	68.9	61.1	50.5	48.3		
		MCWB	43.9	43.2	45.6	50.8	57.7	62.5	65.2	64.3	59.9	55.2	48.2	46.1		
	10%	DB	44.2	43.3	48.6	57.0	67.8	71.6	75.6	74.3	65.3	58.0	48.3	45.8		
		MCWB	41.7	40.4	44.7	48.8	56.3	60.1	63.6	62.9	58.0	53.3	46.4	44.0		
	0.4%	WB	49.0	51.5	52.5	57.6	64.3	69.7	70.2	70.0	66.3	62.1	52.3	50.3		
		MCDB	51.0	56.2	60.8	69.4	76.8	80.9	81.4	83.3	75.4	68.9	54.2	51.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.4	47.2	49.6	54.1	61.6	66.0	68.0	67.2	63.8	58.7	50.4	48.5		
		MCDB	48.3	50.8	54.5	63.5	72.8	76.9	79.7	78.0	71.3	64.4	52.3	50.5		
	5%	WB	44.1	43.5	47.1	51.8	59.0	63.5	66.0	65.5	61.3	55.8	48.5	46.3		
		MCDB	46.3	46.4	50.7	59.5	69.9	73.7	77.6	75.4	67.7	59.8	50.2	48.0		
	10%	WB	42.0	40.5	44.8	49.7	56.8	61.1	64.0	63.8	59.0	53.8	46.4	43.8		
		MCDB	44.1	42.8	48.3	56.3	66.4	70.2	74.2	72.3	64.0	57.6	48.1	45.7		
	5% DB	MDBR	8.3	9.9	11.7	15.5	18.7	17.2	17.9	17.2	14.8	12.4	8.3	7.1		
		MCDBR	8.7	11.9	16.7	22.1	27.0	26.1	25.9	24.3	21.5	16.7	10.5	8.6		
		MCWBR	8.0	9.6	11.3	12.6	13.9	13.2	12.6	11.7	12.6	10.9	8.9	8.2		
		MCWBR	8.4	11.6	15.0	20.5	23.7	23.0	23.3	20.4	19.1	14.9	10.0	8.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.1	9.8	10.9	12.2	13.0	12.5	12.5	10.7	12.3	10.6	8.9	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.336	0.371	0.407	0.403	0.401	0.420	0.414	0.380	0.358	0.334	0.307		
	taud		2.329	2.323	2.258	2.213	2.269	2.295	2.259	2.288	2.363	2.399	2.378	2.326		
	Ebn at Noon		208	240	255	260	268	270	261	256	250	230	199	186		
	Edh at Noon		19	26	34	40	40	39	40	37	30	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		189	404	771	1276	1595	1685	1579	1326	971	532	231	138		
	RadStd		22	56	81	174	196	131	224	106	108	66	25	10		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A

Nomenclature: See separate page